Introductory Mechanics

Assignment:1, Due date: 16-10-2026 at 10:00 AM

Note: Late submission will not consider towards final grade.

Question.1: The position of an object moving along an x axis is given by $x = 3t - 4t^2 + t^3$, where x is in meters and t in seconds. Find the position of the object at the following values of t : (a)1s (e) What is the object's (b) 2s, (c) 3s, and (d)4s displacement between $t = 0$ and $t = 4$s? (f) What is its average velocity for the time interval from $t = 2$s to $t = 4$s?

Question.2: An electron moving along the x axis has a position given by $x = 16te^{-t}$m, where t is in seconds. How far is the electron from the origin when it momentarily stops?

Question.3: A muon (an elementary particle) enters a region with a speed of 5.00×10^6m/s and then is slowed at the rate of 1.25×10^{14}m/s^2. (a) How far does the muon take to stop?

Question.4: Problem # 17 of text book.

Question.5: Sample problem # 2.05 (Time for full up-down flight, baseball toss) of text book.